

Gabriel Guimarães Carvalho

Research Scientist | Computer Vision | Machine Learning | Ph.D. Physics

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Summary

Hi, I'm Gabriel. I'm a physics PhD who enjoys turning mathematics, scientific computing, computer vision, and AI into intuitive solutions for understanding visual data. I'm also fascinated by the future of computing, especially quantum technologies and generative AI.

Education

Università di Roma "La Sapienza" <i>Ph.D. in Physics and Relativistic Astrophysics</i>	2014 – 2017 Rome, IT
Universidade Federal de Pernambuco (UFPE) <i>M.Sc. in Mathematics</i>	2011 – 2013 Recife, BR
Universidade Federal de Pernambuco (UFPE) <i>B.Sc. in Mathematics</i>	2008 – 2010 Recife, BR

Experience

Research Scientist <i>imec</i>	Jun. 2024 – Present Antwerp, BE	Postdoctoral Researcher <i>UFRPE</i>	2017 – 2018 Recife, BR
Deep Learning - Computer Vision - Genetic Algorithms - R&D in several domains		Alternative gravitational theories and geometric methods in Physics	
Senior Research Scientist <i>IDLab - UAntwerpen - imec</i>	Jan. 2024 – May 2024 Antwerp, BE	Lecturer <i>UFPE</i>	Nov. 2017 – Jul. 2019 Recife, BR
Deep Learning - Computer Vision - R&D		Calculus I, II, III - Analytic Geometry - Linear Algebra - Calculus w/ One Complex Variable	
Computer Vision Engineer <i>SpaceTime Labs & Moray</i>	Jul. 2021 – Nov. 2023 São Paulo, BR (Remote)	Lecturer <i>UFPE</i>	Apr. 2012 – Jan. 2014 Recife, BR
Computer Vision - Image Processing - Data Preprocessing/Analysis - Hyperspectral Imaging - Geographic Information System (GIS)		Calculus I, II - Analytic Geometry	
Ph.D. Researcher <i>Projeto CIn - Motorola</i>	Sep. 2019 – Jun. 2021 Recife, BR		
Computational Photography - Deep Learning - Data Science/Analysis - Image Processing - Image QA			

Skills

Languages: Proficient: Python. Familiar: Go, Matlab, C++, Java	DevOps: Git, Docker, AWS. Familiar: MLFlow, DVC
ML/CV: PyTorch, PyTorch Lightning, OpenCV, Scikit-Learn, Scikit-Image, NumPy, SciPy, Pandas	OS: Linux, macOS, Windows
Geospatial: QGIS, GeoPandas, Rasterio	Spoken: Portuguese (native), English (C1/C2), Italian (B2/C1), Norwegian (A1/A2)

Certifications

IBM DevOps & Software Engineering · Deep Learning (DeepLearning.AI) · Math for ML (Imperial College) · ML (Stanford)
Advanced CV with TensorFlow · Quantum Information



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